

Recovering the unwritten recurrences of the table OEIS A253544

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Abstract

OEIS A253544 is a two-dimensional table $T(n, k)$ whose empirical recurrences are, for several of its lines, given only as an ORDER — “k=2: [order 19]” and the like — with the coefficients in a linked file rather than in the entry. Those conjectures can still be settled, and the recurrences recovered. A line of the table is a fixed-width or fixed-height array count, hence a walk count on finitely many vertices, hence satisfies some monic recurrence of order at most that number; Berlekamp–Massey on twice as many exact terms returns the MINIMAL one. Where its order equals the order the entry states — and where the same holds after the first few terms are dropped — any recurrence of that order the line satisfies has a characteristic polynomial that is a multiple of the minimal one and of equal degree, hence is that polynomial. This note settles column 2, column 3.

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1 The table and the unwritten conjectures

OEIS A253544 is “ $T(n, k)$ =Number of $(n+2) \times (k+2)$ 0..1 arrays with every 3×3 subblock diagonal maximum minus antidiagonal minimum nonincreasing horizontally and nondecreasing vertically.”. It is read by antidiagonals and begins

512, 3328, 3328, 20480, 33568, 20480, 119040, 317696, ...

The entry carries, among its empirical claims:

k=2: [order 19]

k=3: [order 21]

As of the “Last modified” line on the live entry (Jul 23 2025 14:07:30, revision 6) these are still recorded as empirical, and the recurrences themselves are not in the entry text.

2 Why a line of the table is a walk count

Fix the index that the line fixes. The entry defines $T(n, k)$ as a number of arrays of a shape determined by n and k , subject to a condition local to a bounded window of consecutive lines. Substituting the fixed index into the entry’s own wording turns the definition into an ordinary one-parameter array count, and for such a count the admissible windows are the vertices of a finite digraph and an array is a walk. So the line is a walk count on S vertices, and by the Cayley–Hamilton theorem it satisfies a monic integer recurrence of order at most S .

3 Recovering the recurrences

Lemma 1. *A sequence known to satisfy some linear recurrence of order at most S is determined, with its minimal recurrence, by its first $2S$ terms; Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Column $k = 2$

Substituting the fixed index into the entry’s wording gives an ordinary one-parameter array count, a walk on $S = 448$ vertices. Berlekamp–Massey on $2S = 896$ exact terms returns a minimal recurrence of order 19 — the order the entry states — with integer coefficients

c_1	13	c_6	1721	c_{11}	266368	c_{16}	24772608
c_2	−37	c_7	−11813	c_{12}	−103680	c_{17}	−31457280
c_3	−95	c_8	86096	c_{13}	1683712	c_{18}	−12582912
c_4	1181	c_9	−17360	c_{14}	−11894784	c_{19}	16777216
c_5	−3897	c_{10}	−280192	c_{15}	12763136		

so that $a(m) = \sum_{i=1}^{19} c_i a(m-i)$ for every m . Removing the first $19 + 4$ terms and repeating gives order 19 again, which is what the identification below needs.

Column $k = 3$

Substituting the fixed index into the entry’s wording gives an ordinary one-parameter array count, a walk on $S = 1792$ vertices. Berlekamp–Massey on $2S = 3584$ exact terms returns a minimal recurrence of order 21 — the order the entry states — with integer coefficients

c_1	17	c_7	261248	c_{13}	46170112	c_{19}	16777216
c_2	−11	c_8	−776832	c_{14}	−132120576	c_{20}	−167772160
c_3	−693	c_9	664576	c_{15}	1835008	c_{21}	67108864
c_4	3244	c_{10}	−1263616	c_{16}	86507520		
c_5	−19932	c_{11}	−15355904	c_{17}	−117440512		
c_6	46560	c_{12}	39215104	c_{18}	176160768		

so that $a(m) = \sum_{i=1}^{21} c_i a(m-i)$ for every m . Removing the first $21 + 4$ terms and repeating gives order 21 again, which is what the identification below needs.

Theorem 2. *For each line above, the displayed recurrence holds for every index, and it is the recurrence the entry records.*

Proof. It holds by Lemma 1. For the identification, the entry asserts that the line satisfies SOME recurrence of the stated order, possibly only from some index on. Let q be its characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail. Then $q_{\min} \mid q$, and the second Berlekamp–Massey run — on the line with its first few terms removed — shows $\deg q_{\min}$ equals the stated order, which is $\deg q$. Both are monic, so $q = q_{\min}$. $\square$

4 Verification

Three checks.

First, each line’s model was compared against the entry’s own published values for that line, taken from the antidiagonal DATA read in either orientation and from the “Table starts” block, and agrees at every available term. This is the only point at which reading the entry’s English enters, and a misreading gives different counts.

Second, each recovered recurrence was evaluated directly on the model’s values, with no Berlekamp–Massey involved, and holds at every index tested.

Third, each order was computed twice by independent routes — modulo a 61-bit prime, where every intermediate is a machine word, and again in exact rational arithmetic — and once more on the line with its first few terms dropped, which is what the identification requires. All agree.

References

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- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
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